

Equilateral Triangle Problem:

Equilateral (~~triangle~~ ABC) $\rightarrow$ isosceles (ABC) ... (i)

isosceles (ABC) $\rightarrow$ equal (AB, AC) ... (ii)

equal (AB, AC) $\rightarrow$ equal (B, C) ... (iii)

Equilateral (ABC) ... (iv)

$\neg$ Equal (B, C) Prove.

Convert the sentence into clausal form:

$\neg$ Equilateral (ABC) $\vee$ isosceles (ABC) ... (i)

$\neg$ Isosceles (ABC) $\vee$ equal (AB, AC) ... (ii)

$\neg$ equal (AB, AC) $\vee$ equal (B, C) ... (iii)

Equilateral (ABC) ... (iv)

$\neg$ Equal (B, C) $\neg$ equal (AB, AC) $\vee$ equal (B, C)

$\neg$ isosceles (ABC) $\vee$ equal (AB, AC) $\neg$ Equal (AB, AC)

$\neg$ isosceles (ABC) $\neg$ equilateral (ABC) $\vee$ isosceles (ABC)

Equilateral (ABC) $\neg$ equilateral (ABC)

$\square$ null clause